

XREAL One Pro Test Protocol v2 — Ambient Council-aligned

XREAL One Pro Test Protocol — Shadow x Yeshiva U (v2, Ambient Council-aligned)

Version: 2.0 · **Prepared:** 2026-07-07 (post 4-agent full-web deep research + design pivot to Ambient Council v3) **Supersedes:** v1.0 (2026-07-06) — which assumed standalone rendering + persona semicircle. Both assumptions were wrong. **Hardware:** XREAL One Pro (\$599, Lora purchased 2026-07-06) + **XREAL Eye add-on (\$99, ORDER NOW)** + Android phone (Pixel 9 Pro / Samsung S24) or MacBook Pro as USB-C tether host **Software backend:** Shadow v1.5.14 (github.com/alex-jb/shadow-mentor, MIT, public, 706/707 tests) **Rendering pipeline:** WebXR + Three.js + gpt-realtime voice (WebRTC) + Perplexity-style citation pills — served over Chrome on the tether host **Design paradigm:** Ambient Council v3 (docs/product-design/xr-visual-paradigm-2026-07-06-v3-AMBIENT-COUNCIL.md)

Purpose: define what the July 2026 Dean + Vice-Provost demo actually looks like, what pass/fail criteria we agree on before the demo date, and what the Alex + Loredana test-week protocol is on-device.

Authors: Alex Xiaoyu Ji (design + implementation) · Loredana C. Levitchi (regulatory-domain review + hardware ownership)

What changed from v1

The v1 protocol had two structural errors baked in that would have wasted Phase B on-device time:

1. **v1 assumed XREAL One Pro is a standalone computer.** It is not. It is a USB-C DisplayPort-Alt-Mode peripheral. iPhone Safari does NOT support WebXR immersive-AR. The correct compute host is Android + Chrome + WebXR, or Mac + Chrome window as a 2D fallback.
2. **v1 assumed a “5 personas floating in tribunal” showcase.** Full-web research on 12 years of shipping bank AR/VR demos (Fidelity StockCity 2014 → Meta Horizon Workrooms killed 2026-02-16) showed every floating-showcase demo died in the day-30 utility test. v3 Ambient Council paradigm replaces this with document-anchored Perplexity-style pills + spatialized voice + Severance restraint.

If you are reading v1 (`git log --diff-filter=D`) — throw it out. Read only v2.

Ownership and coordination

- **Loredana** owns the physical device + XREAL Eye purchase + regulatory–domain scene review + executive coordination with the Dean’s office.
- **Alex** owns the software pipeline: Shadow API + WebXR scene + gpt–realtime voice + Perplexity citation pills.
- **Both** sign off in writing before the Dean + VP demo on each scene’s readiness. No solo approvals.

Hardware pipeline (corrected)

The critical purchase (do this today, ~\$99)

XREAL Eye add-on — 12MP forward camera + native 6DoF spatial anchoring on the X1 chip. General availability since July 2025. Ships from us.shop.xreal.com/products/xreal-eye in 2–4 business days. Without the Eye, XREAL One Pro is display–only with no camera and no 6DoF — Ambient Council’s document–anchored HUD cannot work.

Total additional cost: \$99.

The compute host (borrow or use MacBook)

- **Primary:** Android phone with USB–C DisplayPort–Alt–Mode support — Pixel 9 Pro, Samsung S24 Ultra, or OnePlus 12. Runs Chrome + WebXR + Three.js + gpt–realtime voice.
- **Fallback:** MacBook Pro (any Apple Silicon). Runs Chrome window that gets cast to XREAL One Pro as a 171–inch 2D virtual screen. Loses head–tracked rendering; retains persona positions and voice interaction as a fixed 2D image.

iPhone is not a host because Safari does not support WebXR immersive–AR (verified via caniuse.com/webxr 2026–07–07). Do not plug XREAL One Pro into an iPhone for the demo.

Voice stack

- **Primary voice model:** gpt–realtime (OpenAI, WebRTC transport, 200–350ms first–audio latency, native barge–in). Reference: [docs/strategy/roadmap-2026-2028.md](#) § voice.
- **Fallback for deictic references** (“this loan”, “that flag”): Gemini 3.1 Flash Live (1M–token multimodal context).
- **Activation:** no wake word. Copy visionOS 27 Siri orb pattern — a gaze–anchored Council orb pinned above the physical loan PDF. Look–and–speak.

What we do NOT buy

- No Meta Quest 3S backup (\$299). VR headset aesthetic breaks the Ambient Council eyeglass framing. Fallback is MacBook + Chrome, not another head-mounted device.
 - No Meta Ray-Ban Display (\$799). Monocular 20° HUD is too small for the 4-element document-anchored layout.
 - No Apple Vision Pro (\$3,499+). Publicly de-prioritized (Kuo 2026-06-03), off-brand for the pitch.
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Phase A — Preflight (2026-07-08 to 2026-07-10, no hardware needed)

While the XREAL Eye is in transit, Alex builds and validates the WebXR scene on desktop Chrome + WebXR emulator.

A1. Ambient Council scene skeleton

Alex builds the 4-element HUD: – Verdict pill (top-right, above document region) – 5 collapsed voice cards (right column, expand on voice command or pinch) – Attestation status pill (bottom-right) – Optional AML/KYC per-line callouts anchored to specific lines of the PDF

Content spec + design references at docs/product-design/xr-visual-paradigm-2026-07-06-v3-AMBIENT-COUNCIL.md.

Voice interaction: user says “explain the compliance voice” → the corresponding voice card expands with the full rationale + Reg B citation pill + AA code. User says “audit trail” → the scene shifts to Joi-style volumetric persona replay for the past 30 days of decisions (reserved for this one use case only).

A2. WebXR emulator dry run

Chrome DevTools → Sensors → Virtual reality profile: – Scene loads without runtime error on Chrome 137+ – 4 HUD elements render in correct positions relative to a mock PDF surface – Verdict pill = green/amber/red based on mock verdict response – Voice command “explain the compliance voice” expands the correct card – Fade-in 300ms / fade-out 500ms on head-turn simulation

A3. iPhone camera + Shadow API round-trip

Two-layer honest demo (per v3 design): – **Layer 1:** iPhone camera + VisionKit VNRecognizeTextRequest on-device OCR + POST /api/deliberate (Shadow API) round-trip. Verify < 2s on Wi-Fi, < 3s on cellular. – **Layer 2:** the XREAL Eye captures the same document + renders the HUD. This is the “target platform” demo layer.

Both layers must work by 2026-07-10 EOD.

Phase B — On-device test (2026-07-14 to 2026-07-18)

Loredana brings XREAL One Pro + Eye to school. 5 sessions, 60 minutes each, one per day.

B1 (Mon 2026-07-14) — Tethered baseline

Boot XREAL One Pro + Eye + Android host or MacBook. Load Chrome + WebXR scene.

Measure:

- Cold load: < 3s from Eye document capture to HUD render
- Sustained FPS: ≥ 45 fps on Android tether, ≥ 60 fps on MacBook fallback, over a 3-minute session with 4 HUD elements + spatialized voice
- HUD anchor drift: < 5mm at 40cm document viewing distance over 30 seconds
- Voice round-trip: < 1.5s p95 from utterance to card expansion

Pass criteria: all four numeric thresholds met on the primary path (Android + Chrome + WebXR). If Android fails on FPS or anchor drift, degrade to MacBook fallback and record the fallback numbers.

B2 (Tue 2026-07-15) — 4-element HUD readability

Same scene, real Shadow API endpoint. Measure:

- Verdict pill readable at 40cm without leaning in
- 5 voice cards expand on voice command in a quiet room (rehearsal room, no noise)
- Attestation status pill (green ✓ / red ✗) legible from the reviewer's seated posture
- Optional AML/KYC per-line callouts: track accuracy on 8pt loan text. If drift > 3mm/callout, DROP the per-line feature and put flags at the top of the Compliance voice card instead.

Pass criteria: verdict + voice-card expand + attestation pill all pass; per-line callouts are gambled — accept the accuracy result whatever it is.

B3 (Wed 2026-07-16) — 4-case verdict lattice

Load each of the 4 canonical case studies (approve × 1, escalate × 1, block × 2 paths) from docs/case-studies/. Run each end-to-end. Measure:

- Verdict correctness against expected verdict (4/4)
- 5 voices render + are voice-callable (5/5)
- Hash-chain audit entry verifies via /api/verify-chain (4/4)
- Voice-command “explain compliance voice” surfaces the correct rationale text with the correct AA code and Reg B / SR 26-2 / CFPB citation

Pass criteria: 4/4 verdict correctness, 5/5 voice availability, 4/4 audit entries verify, all voice-invoked drill-downs surface the right rationale.

B4 (Thu 2026-07-17) — Executive demo dress rehearsal

Full 5-minute Dean + VP demo rehearsal. The Ambient Council paradigm keeps this shorter than v1 (was 8 min; now 5 min):

- Open scene from tethered laptop or Android phone (30s)
- Play Case 1 (\$2.5M CRE loan) — 90s
- Reviewer says “explain the compliance voice” → HUD updates → 45s
- Reviewer says “audit trail” → scene shifts to Joi-style volumetric replay for 30 days → 60s
- Reviewer says “back to loan” → return to Case 1 HUD → 30s
- Q&A anticipation (10 min prep — off the demo clock)

Pass criteria: end-to-end demo runs in ≤ 5 minutes, both Alex and Loredana can narrate confidently.

B5 (Fri 2026-07-18) — Failure mode red-team

Deliberately break 5 things + measure recovery:

1. Kill Wi-Fi mid-verdict → does the client retry gracefully or hang?
2. Send a tampered attestation → does client-side verify catch it?
3. Trigger WebXR permission revoke mid-session → does scene degrade to 2D?
4. Speak “explain the compliance voice” while another voice is expanded → does the interruption pattern (gpt-realtime barge-in) actually work?
5. Occlude the XREAL Eye camera for 3 seconds → does the HUD gracefully hide + re-appear on re-detect?

Pass criteria: all 5 failures recover gracefully. None crash the browser or freeze the scene.

Phase C — Dean + Vice-Provost executive demo (target: last week of July)

- Loredana coordinates the date with the Dean’s office.
- Alex + Loredana demo Case 1 (\$2.5M CRE loan) as the primary narrative.
- Backup: Case 3 (SBA loan, OFAC SDN hit → block by AML/KYC) if reviewer asks for a second scenario.
- Fallback path if XREAL fails on the day: MacBook + Chrome window running same WebXR scene as 2D preview. Loses head-tracked rendering; retains everything else.
- Loom recording (unlisted) of the actual session in Loredana’s Drive within 24h.
- Post-demo debrief: agree on what publishes vs stays private.

Failure protocol

If any Phase B session fails a pass criterion: 1. Log in docs/xreal-one-pro-test-protocol/failure-log.md 2. Alex has 24h to iterate the scene / endpoint / fallback 3. Rerun the failed session within 48h 4. Do NOT proceed to Phase C until every Phase B session is green

If a systemic hardware failure (X1 driver crash, XREAL SDK regression, USB-C DP-Alt loss): – Alex contacts XREAL developer support – Loredana communicates the slip to the Dean's office – Fall back to MacBook + Chrome window as the demo target — the Ambient Council WebXR scene works identically here, just as 2D not head-tracked.

Contact

- Alex Xiaoyu Ji · xji1@mail.yu.edu — software pipeline + WebXR scene + voice stack
 - Loredana C. Levitchi · [email verify] — hardware ownership + Eye add-on purchase + executive coordination + regulatory-domain scene review
 - Jason Marsh · jason@flow.gl — Flow Immersive template support (fallback layer only, not primary path)
 - Hieu Ngo, PhD · Yeshiva Katz School — XR faculty consult, IEEE VR co-author
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This protocol is a public artifact of shadow-mentor (MIT). Every design decision above is grounded in either v3 Ambient Council design doc, 4-agent full-web deep research 2026-07-06 through 2026-07-07, or the Strategic Roadmap 2026-2028 doc. Open an issue if any claim is out of date.